

Project Report

1. INTRODUCTION

1.1 Project Overview

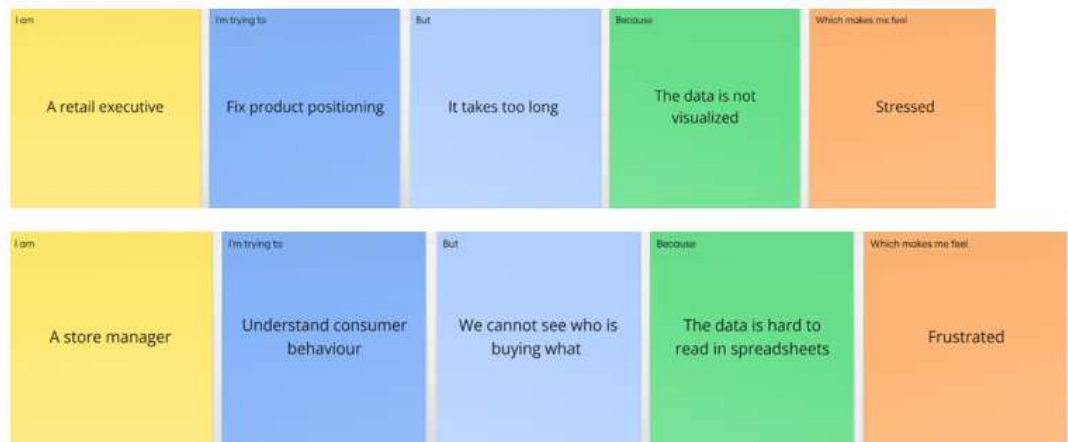
This project aims to investigate the relationship between product positioning, sales performance, and consumer behavior. Using Tableau, we will analyze data to uncover insights into how different positioning strategies impact sales and consumer preferences. By visualizing the data, we aim to provide actionable recommendations to optimize product positioning strategies and drive revenue growth. A retail company wants to understand the impact of product positioning on its sales and consumer behavior. They have collected data on sales figures, product placement, and consumer demographics. They seek insights into which product positioning strategies are most effective in driving sales and how they can tailor their marketing efforts accordingly. Through data visualization with Tableau, the company hopes to gain actionable insights to improve its product positioning strategies and increase revenue.

1.2 Purpose

The primary objective of this project is to investigate the direct relationship between strategic product positioning, sales performance, and consumer behavior within a retail environment. By leveraging transactional records, product placement schemas, and customer demographic data, this analysis utilizes Tableau to transform raw retail data into an interactive visual asset. Ultimately, the project seeks to identify high-performing store locations, understand how different consumer segments interact with physical layouts, and uncover exactly how specific positioning strategies influence overall sales lift. The final dashboard will deliver data-driven, actionable recommendations to optimize shelf layouts, maximize spatial revenue efficiency, and accelerate bottom-line business growth.

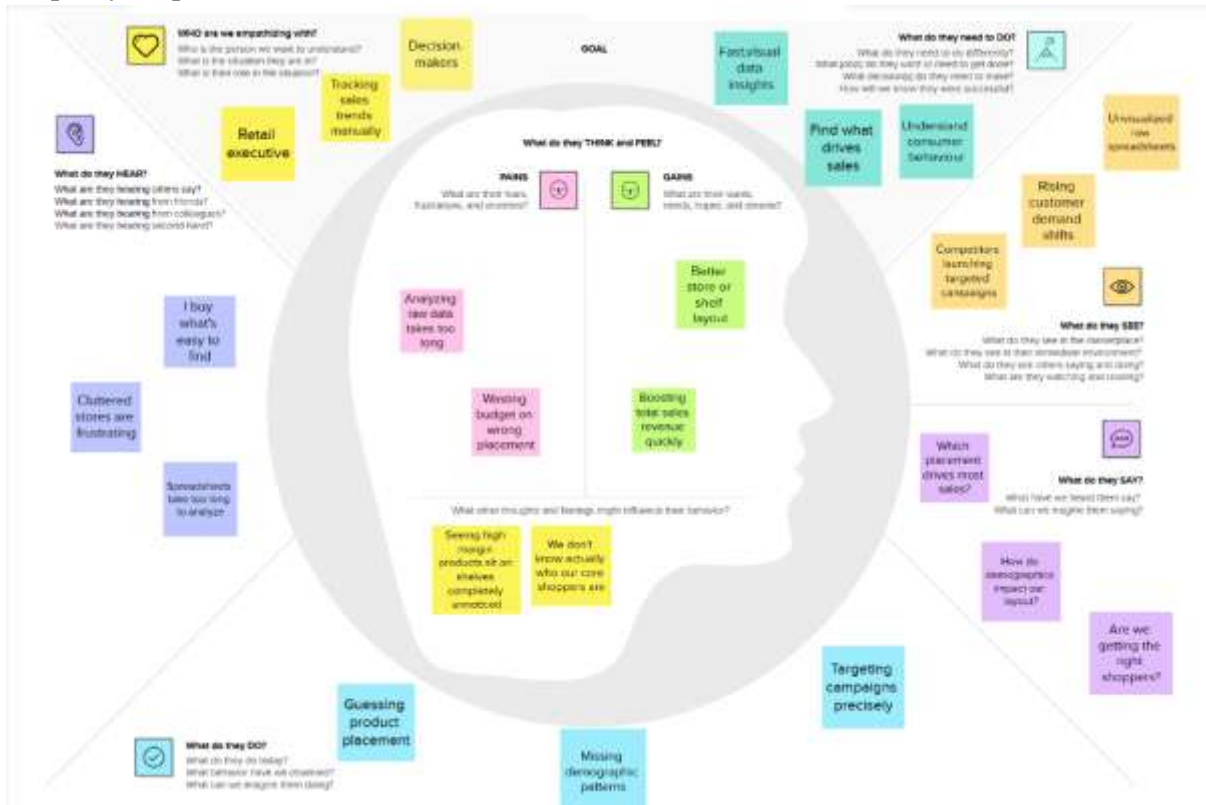
2. IDEATION PHASE

2.1 Problem Statement



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A retail executive	Fix product positioning	It takes too long	The data is not visualized	Stressed
PS-2	A store manager	Understand consumer behaviour	We cannot see who is buying what	The data is hard to read in spreadsheets	Frustrated

2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

- 40 minutes to prepare
- 1 hour to collaborate
- 2-3 people recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

- 10 minutes**
- Team gathering**
Define who should participate in the session and what to make. Share relevant information in advance.
- Set the goal**
Think about the problem you'll be focusing on solving in the brainstorming session.
- Learn how to use the facilitation tools**
Use the Facilitator Superpowers to run a happy and productive session.

[Open article](#)

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

10 minutes

Problem

How might we maximize sales revenue by matching product placements to demographics?

Key rules of brainstorming

To our own concept and productive solution

- Stay on topic
- Encourage wild ideas
- Defer judgement
- Listen to others
- Go for volume
- If possible, one person

Step-2: Brainstorm, Idea Listing and Grouping

Brainstorm

Write down any ideas that come to mind that address your problem statement.

Tip Start with a discussion about the need, context, and big questions first.

Group Ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sort-of code label. If a cluster is large, then do sticky notes, try and see if you can break it up into smaller sub-groups.

Tip Ask yourself: "What does this mean?" "What does this mean?" "What does this mean?"

Revenue Driven placement

place high-margin candy and toys at child-eye level in checkout lanes

Place high-volume staple items (like milk or bread) at the very back of the store to force shoppers to walk past more products

pair related items (like chips and dips) together on aisle end-caps

Step-3: Idea Prioritization

1 Prioritize

Your team should all be on the same page about what's important moving forward. Place your items on this grid to determine which items are important and which are feasible.

[Get solutions](#)

TIP

Use a sticky note that says "I wish I could do this" or "I wish I could do that" to help you decide what to place on the grid. This can also be a helpful way to get feedback from others on what you think is important.

2 After you collaborate

You can export this board as an image or pdf to share with members of your company who might find it helpful.

Quick start-one

- Step 1: Set up the board**
Share a sticky note to the board to get started. You can also use the "Add Item" button to add items to the board.
- Step 2: Organize the board**
Drag items to the "Important" or "Feasible" column to organize the board. You can also use the "Filter" button to filter items by column.

Reorganizing forward

3 Strategy snapshot

Use this board to capture your strategy. You can use the "Add Item" button to add items to the board. You can also use the "Filter" button to filter items by column.

4 Customer experience journey board

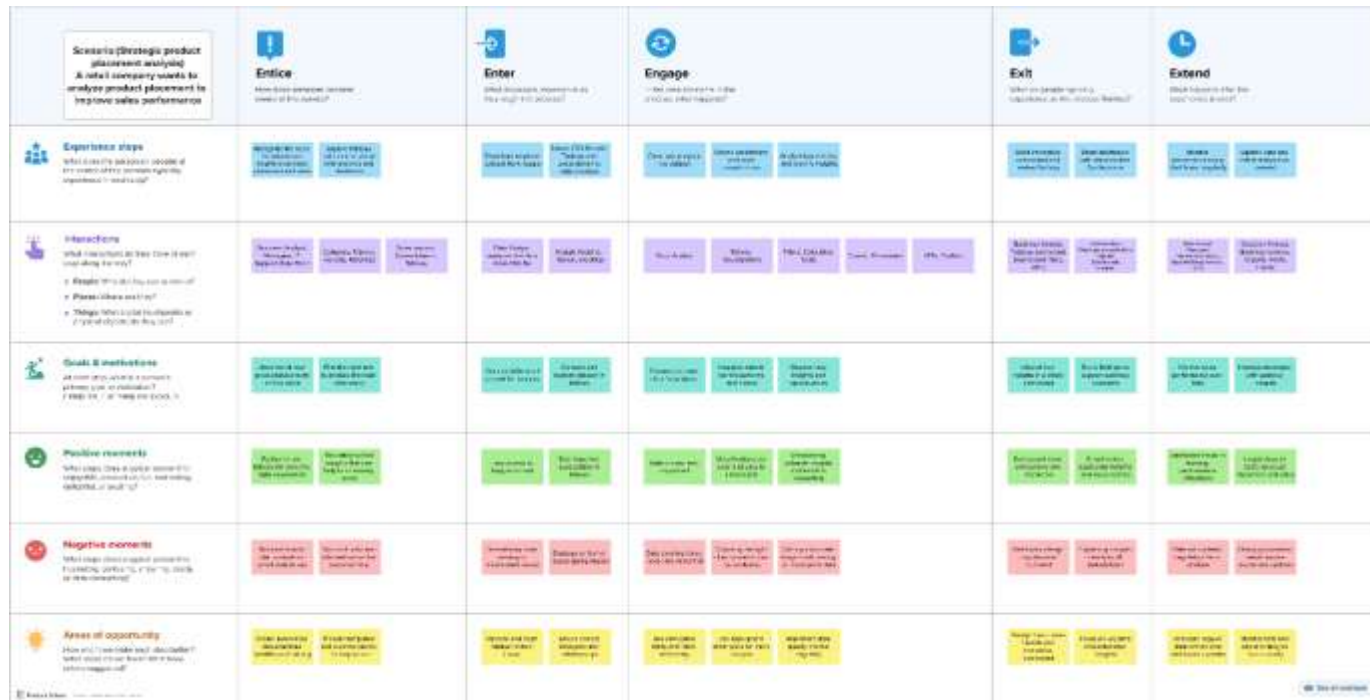
Use this board to capture your customer experience journey. You can use the "Add Item" button to add items to the board. You can also use the "Filter" button to filter items by column.

5 Strategic roadmap, initiatives & ideas

Use this board to capture your strategic roadmap, initiatives, and ideas. You can use the "Add Item" button to add items to the board. You can also use the "Filter" button to filter items by column.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Calculation and KPI metrics	Creating calculated fields. Creating high-level scorecard blocks displaying core metrics with current targets against targets
FR-2	Generate interactive dashboard	Creating unique visualization including KPIs and filters
FR-3	Interactivity and filtering	Creating global filters, multiselect dropdowns
FR-4	Tooltips	Dynamic pop-up text boxes that appear when hovering over datapoints to provide exact metrics

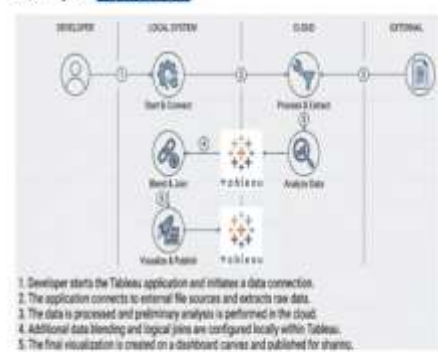
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

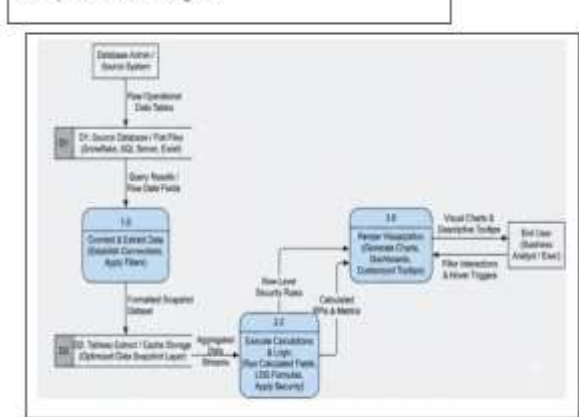
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard interface must be intuitive, easy to navigate.
NFR-2	Security	Data access must be strictly restricted to authorized users via proper authentication
NFR-3	Reliability	Generated reports must consistently deliver accurate and error-free metrics
NFR-4	Performance	The interactive dashboard must fully load and render its visual elements within a few seconds
NFR-5	Availability	The analytics solution must remain reliably accessible to business users whenever required
NFR-6	Scalability	The platform must seamlessly support a growing volume of concurrent users and expanding dataset

3.3 Data Flow Diagram

Example: (Simplified)



Example: Data Flow Diagram



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Analyst	Data Collection	USN-1	Gathering the required dataset from the data source	Dataset is successfully downloaded from Kaggle	High	Sprint-1
	Data Import	USN-2	Import dataset into tableau	Dataset is successfully imported to tableau	Medium	Sprint-1
	Data preparation	USN-3	Clean and validate the dataset	Dataset is cleaned, validated and ready for analysis with no missing or inconsistent values	Medium	Sprint-1
		USN-4	Checking datatypes and creating fields	All columns have the correct data types	Medium	Sprint-1
	Data Visualization	USN-5	Developing visualizations	Visualizations are created for analysis	High	Sprint-2
	Dashboard	USN-6	Developing Dashboard	Interactive dashboard combines all required visualizations successfully	High	Sprint-2
	Story	USN-7	Developing Story	Story presents analysis in logical sequence	High	Sprint-2

3.4 Technology Stack



Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboards for viewing sales and placement insights	Tableau public, Tableau Desktop
2.	Application Logic-1	Data Cleaning and preprocessing	Tableau public
3.	Application Logic-2	Sales and consumer behaviour analysis	Tableau public
4.	Application Logic-3	Product placement analysis and KPI calculations	Tableau public
5.	Database	Data Type, Configurations etc.	None
6.	Cloud Database	Database Service on Cloud	Tableau
7.	File Storage	File storage requirements	CSV file/Excel file
8.	External API-1	Purpose of External API used in the application	None
9.	External API-2	Purpose of External API used in the application	None
10.	Machine Learning Model	Purpose of Machine Learning Model	None
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration; Cloud Server Configuration..	None

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	None
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	None
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	None
4.	Availability	Keep dashboards accessible and backed up	Tableau Public
5.	Performance	Fast data retrieval	Tableau

4. PROJECT DESIGN

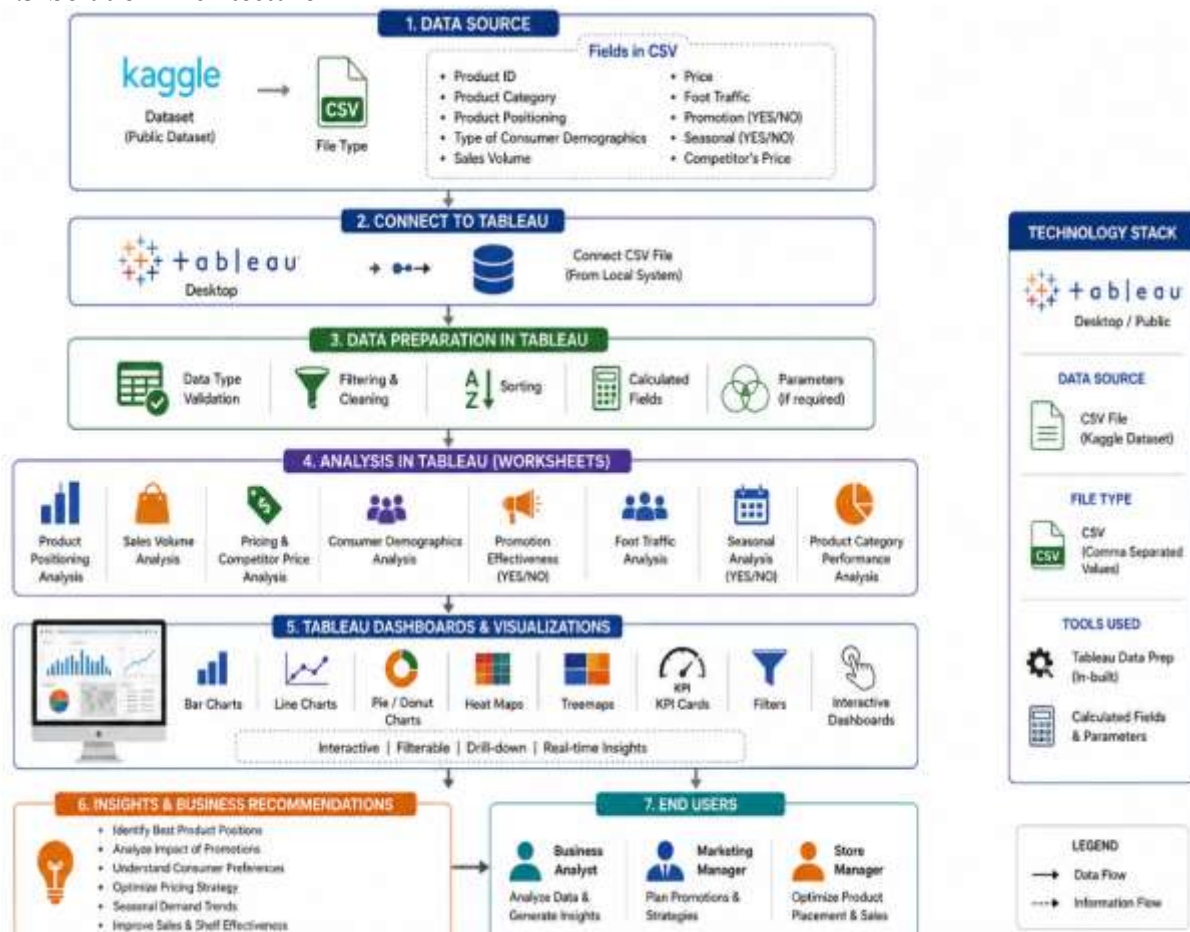
4.1 Problem Solution Fit

1. CUSTOMER SEGMENT(S) Why is your customer? Retail company CS	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action to find their chosen solution? Limited analytical tools, large datasets, time-consuming manual analysis, lack of real-time insights CC	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they have the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? Manual sales reports, Excel spreadsheets, traditional business reports, manager experience AS
2. JOBS-TO-BE-DONE / PROBLEMS Which job(s) need to be done (or problem(s) do you address for your customer)? Analyze product placement, identify best selling locations, understand customer buying behaviour J&P	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? Product placement decisions are based on numerous factors instead of data-driven insights, leading to inefficient shelf utilisation and lower sales RC	7. BEHAVIOUR What does your customer do to address the problem and get the job done? Collect sales data, review reports, compare product performance, analyse customer demographics, exchange shelf reports BE
3. TRIGGERS What triggers customers to act? Increasing sales, poor product visibility, changing customer preferences, demand for better driven decisions TR	10. YOUR SOLUTION Develop a Tableau-based Strategic Product Placement Analytics Dashboard to monitor sales, product placement, and consumer demographics, providing actionable recommendations to optimize product positioning and increase revenue SL	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Address Dashboards, Company portal, Social Reports, Email Reports CH
4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? Before: Confused, overwhelmed product placement After: Confident, informed, able to make better business decisions EM		8.2 OFFLINE What kind of actions do customers take offline? Store visits, Sales meetings, Marketing discussions, Retail audits CH

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Retail companies lack data-driven insights to optimize product placement, resulting in lower sales and inefficient shelf utilization
2.	Idea / Solution description	Develop a tableau-based dashboard that analyses sales, product placement, and consumer demographics to recommend effective product positioning strategies
3.	Novelty / Uniqueness	Combines sales, product placement, and consumer demographics into a single interactive dashboard for data-driven decision making
4.	Social Impact / Customer Satisfaction	Improves shopping experience, help businesses make better decisions, and increases customer satisfaction through optimized product placement
5.	Business Model (Revenue Model)	Subscription based analytics dashboard or licensing solution for retail companies
6.	Scalability of the Solution	Can be implemented across multiple stores, retail chains, and e-commerce platforms with support for large datasets

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Gather dataset from Kaggle	2	High	
Sprint-1	Data Import	USN-2	Import Kaggle CSV file into tableau	1	Medium	
Sprint-1	Data Preparation	USN-3	Clean and validate the dataset	3	High	
Sprint-1	Data Preparation	USN-4	Checking Datatypes	3	Medium	
Sprint-1	Data Preparation	USN-5	Creating calculated fields	3	Low	
Sprint-2	Data Visualization	USN-6	Side-by-side Bar Chart	2	Medium	
Sprint-2	Data Visualization	USN-7	Stacked Bar Chart	2	Medium	
Sprint-2	Data Visualization	USN-8	Scatter Plot	2	Medium	
Sprint-2	Data Visualization	USN-9	Horizontal Bar Chart	2	Medium	
Sprint-2	Data Visualization	USN-10	Heap Map	2	Medium	
Sprint-2	Data Visualization	USN-11	Grouped Bar Chart	2	Medium	
Sprint-2	Data Visualization	USN-12	Tree map	2	Medium	
Sprint-2	Data Visualization	USN-13	KPI	2	Medium	
Sprint-2	Dashboard	USN-14	Developing Dashboard	5	High	
Sprint-2	Story	USN-15	Developing Story	5	High	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	12	5 Days	24 June 2026	28 June 2026	12	28 June 2026
Sprint-2	26	6 Days	29 June 2026	04 July 2026	26	04 July 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$\text{Velocity} = \text{Total Story Points Completed} / \text{Number of Sprints}$$

$$\text{Velocity} = 38 / 2$$

$$19 (\text{Story Points Per Sprint})$$

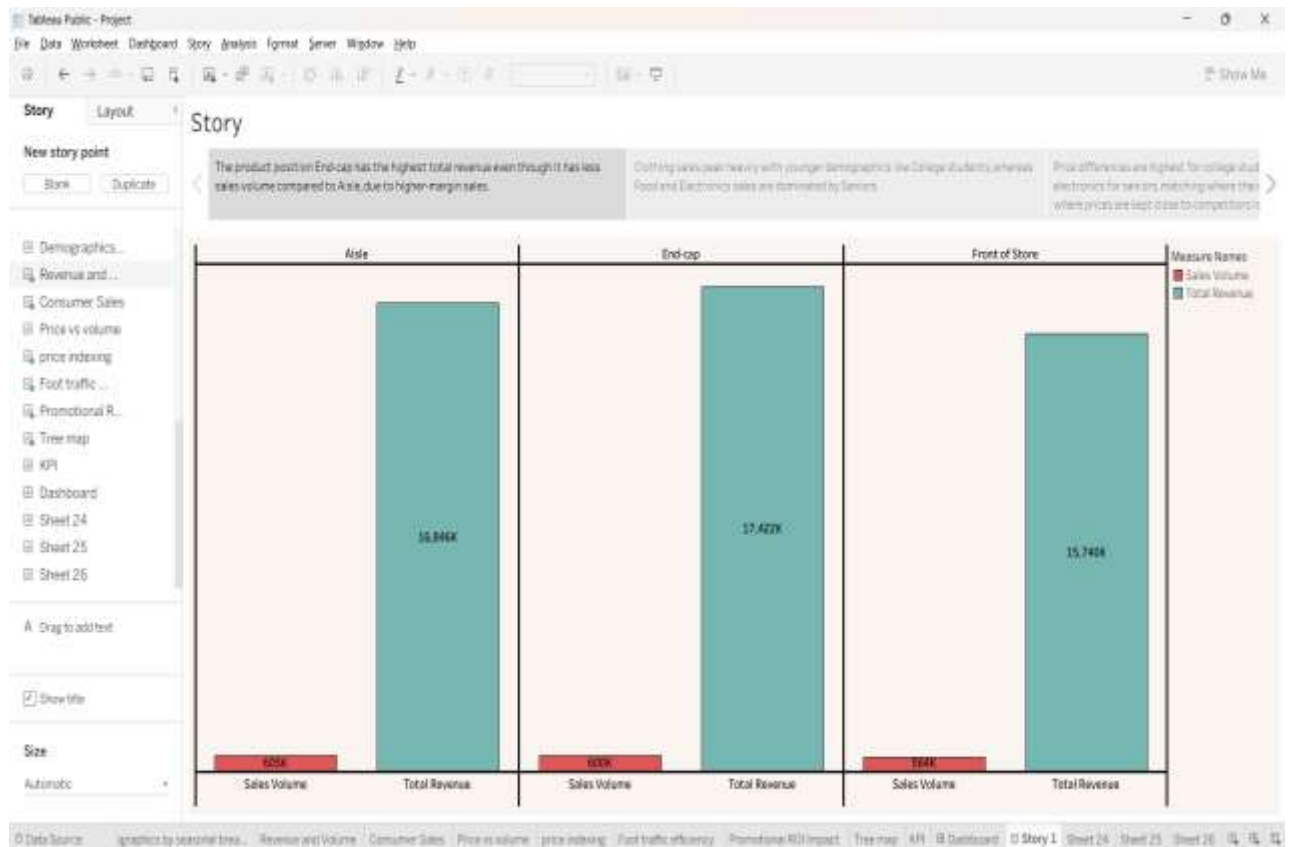
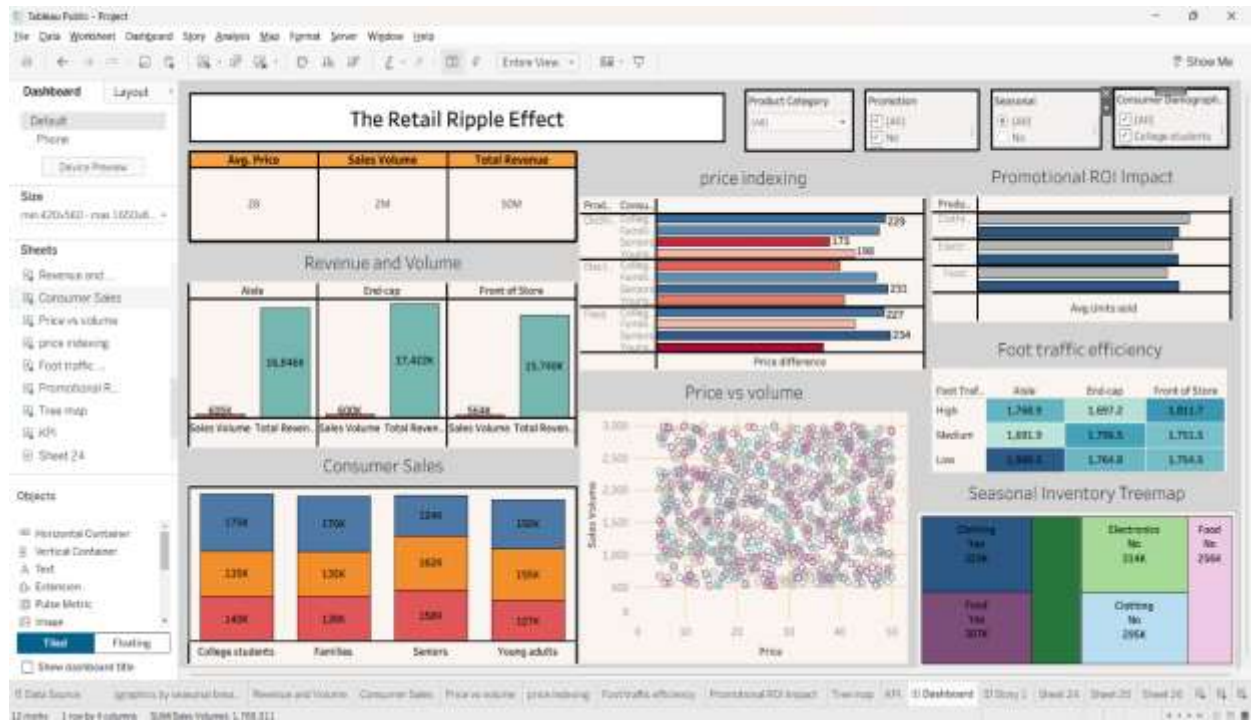
6. FUNCTIONAL AND PERFORMANCE TESTING

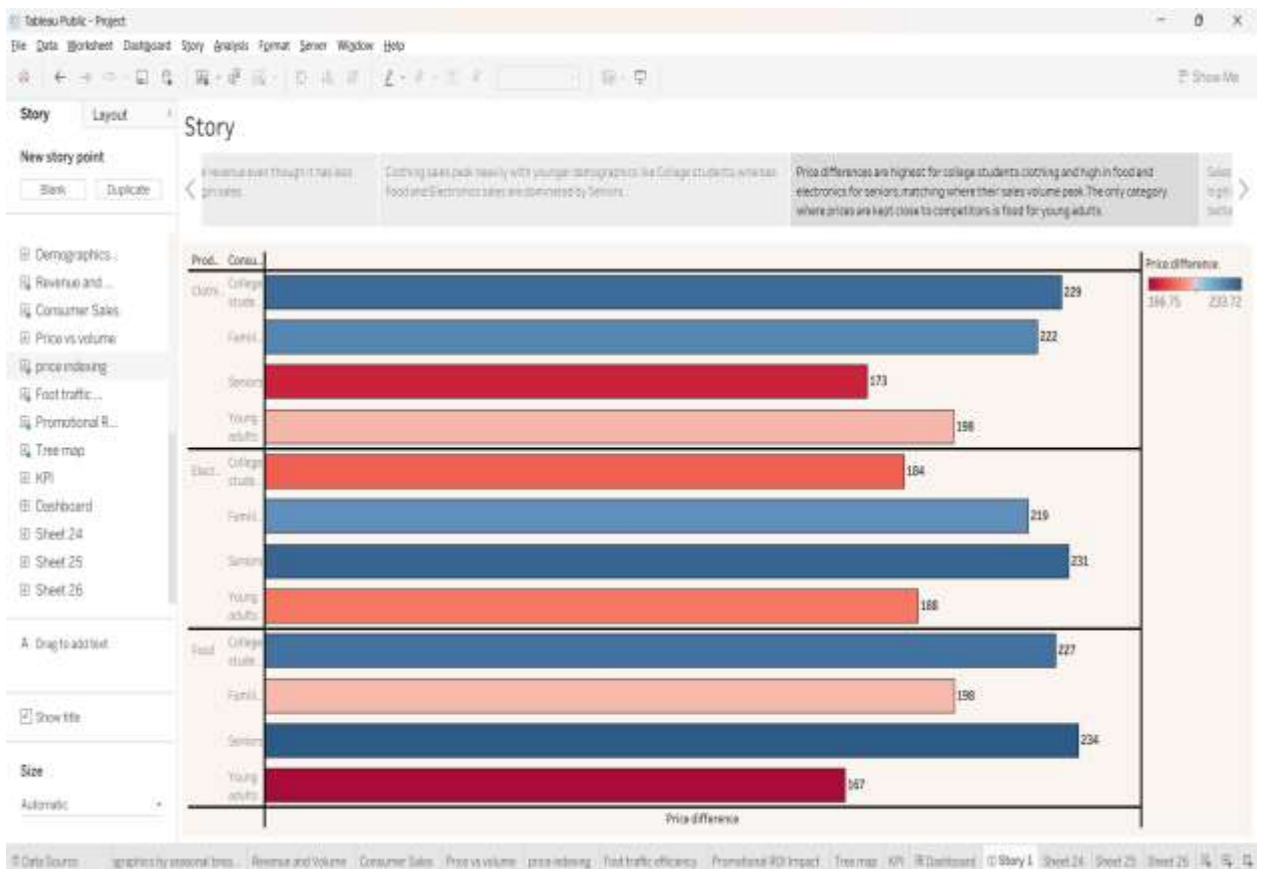
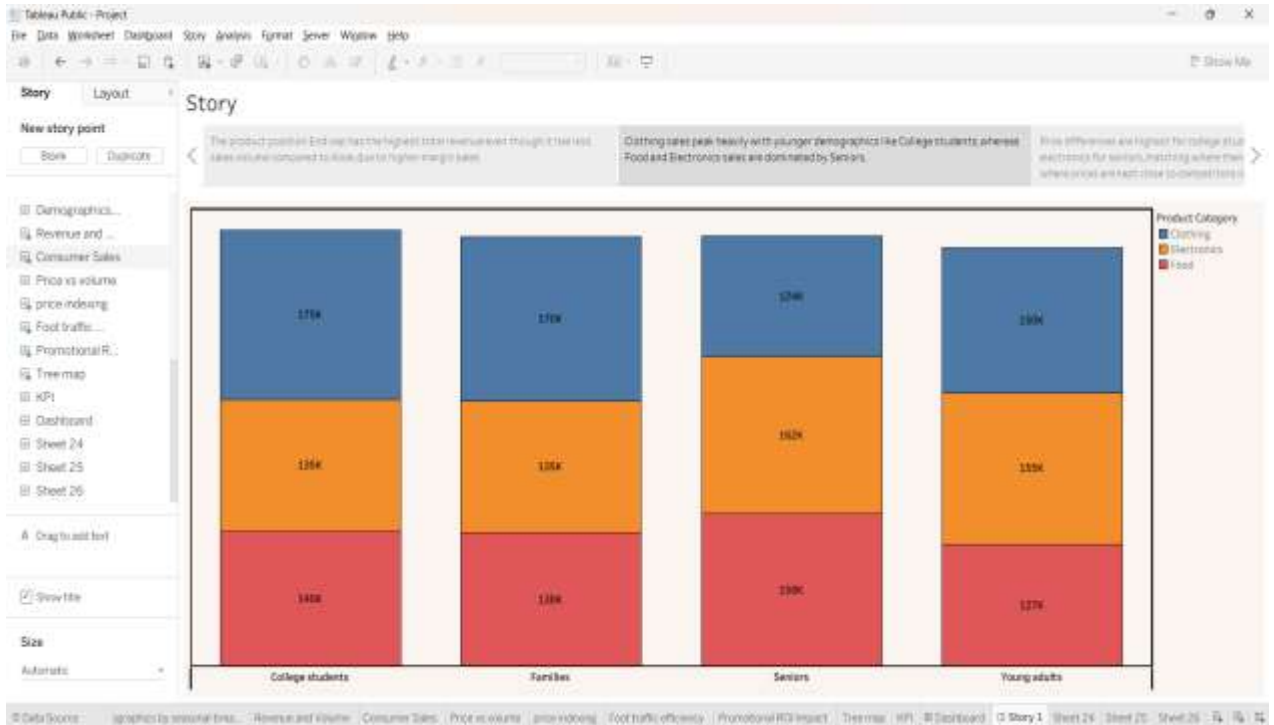
6.1 Performance Testing

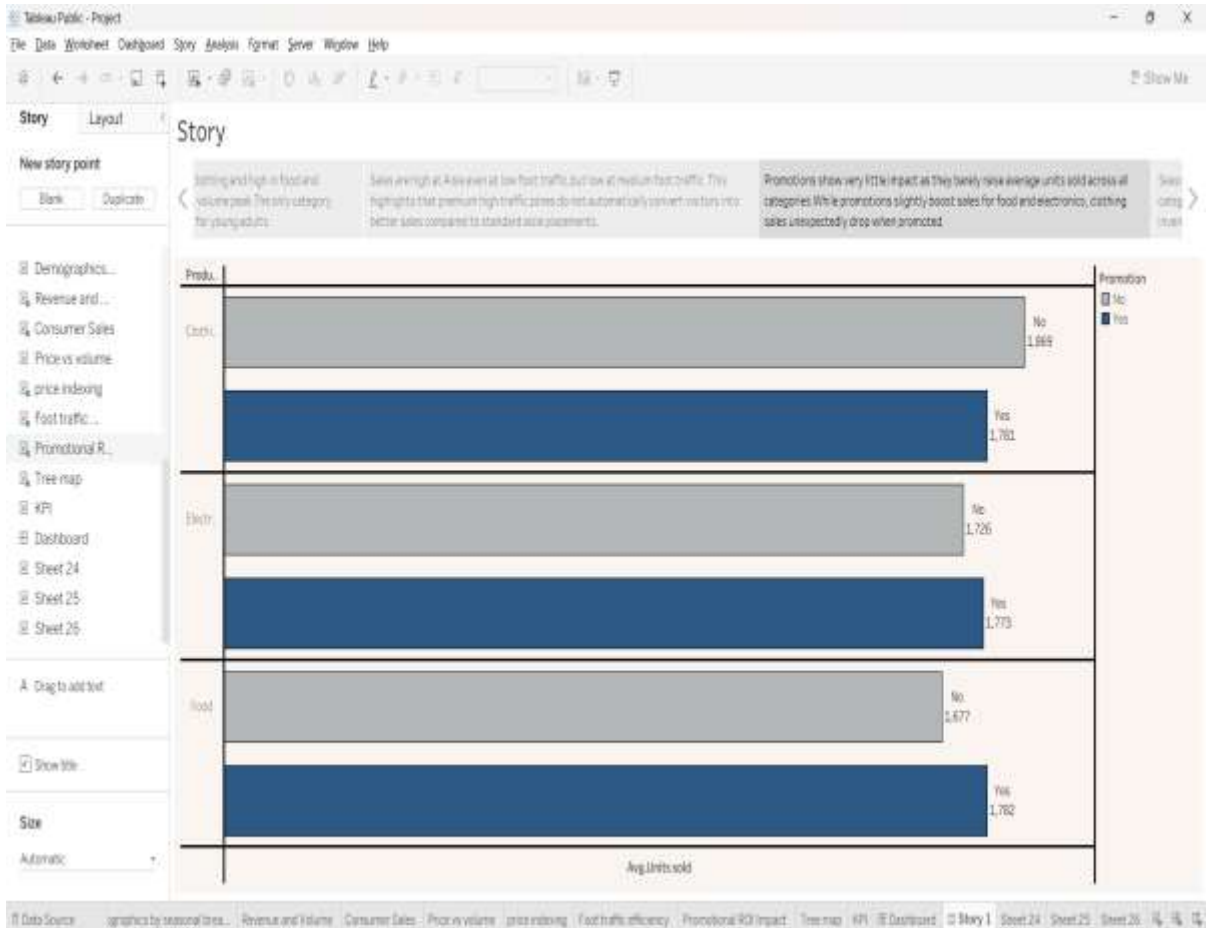
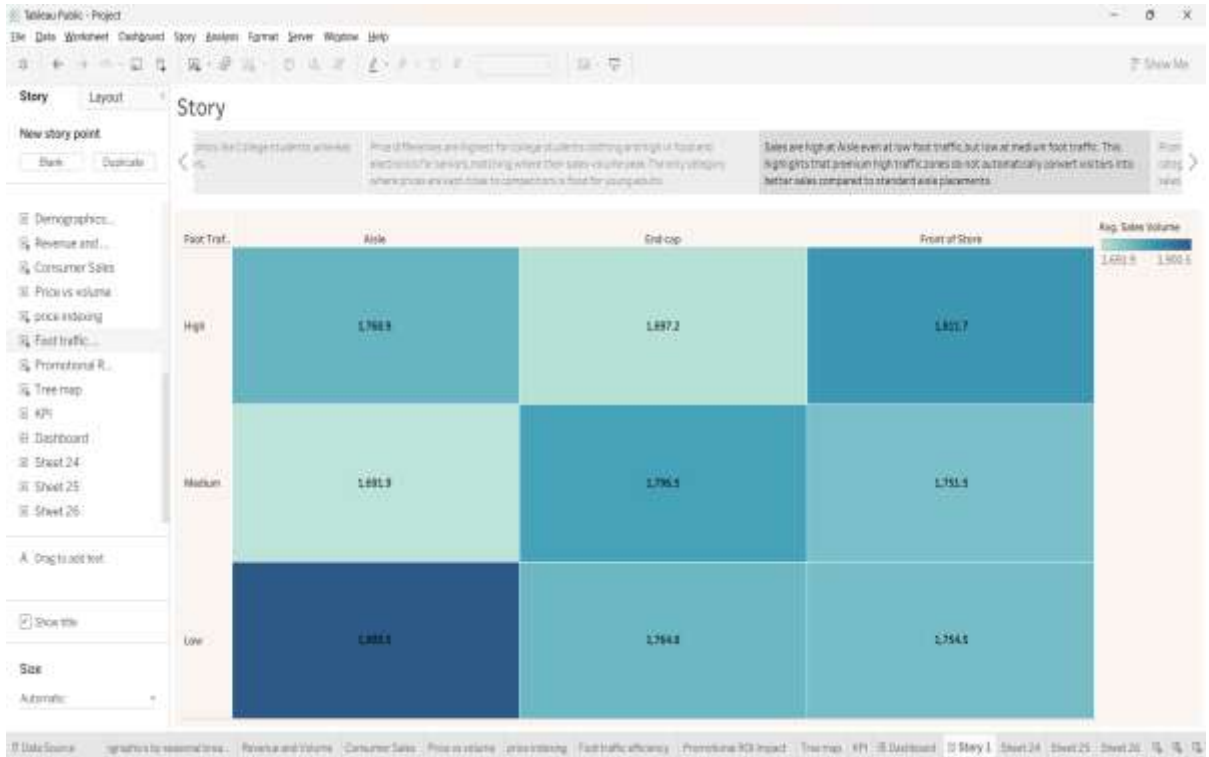
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	CSV File Product positioning Size-66KB 1000 records 10 columns 1001 rows
2.	Data Preprocessing	Cleaned and validated the dataset. Created 2 calculated fields.
3.	Utilization of Filters	4 Filters used: Product category Demographics Seasonal Promotion
4.	Calculation fields Used	2 Calculated fields used: Price Difference Syntax: price-competitor's price Total revenue Syntax: Sales volume*price
5.	Dashboard design	No of Visualizations / Graphs -8 1.Revenue vs Sales volume 2.Consumer Sales 3.Price vs Sales volume 4.Price indexing 5.Promotional ROI impact 6.Foot Traffic Efficiency 7.Seasonal Inventory <u>Treemap</u> .
6	Story Design	No of Visualizations / Graphs -6 1.Revenue vs Sales volume 2.Consumer Sales 3.Price indexing 4.Foot Traffic Efficiency 5.Promotional ROI Impact 6.Seasonal Inventory <u>Treemap</u> .

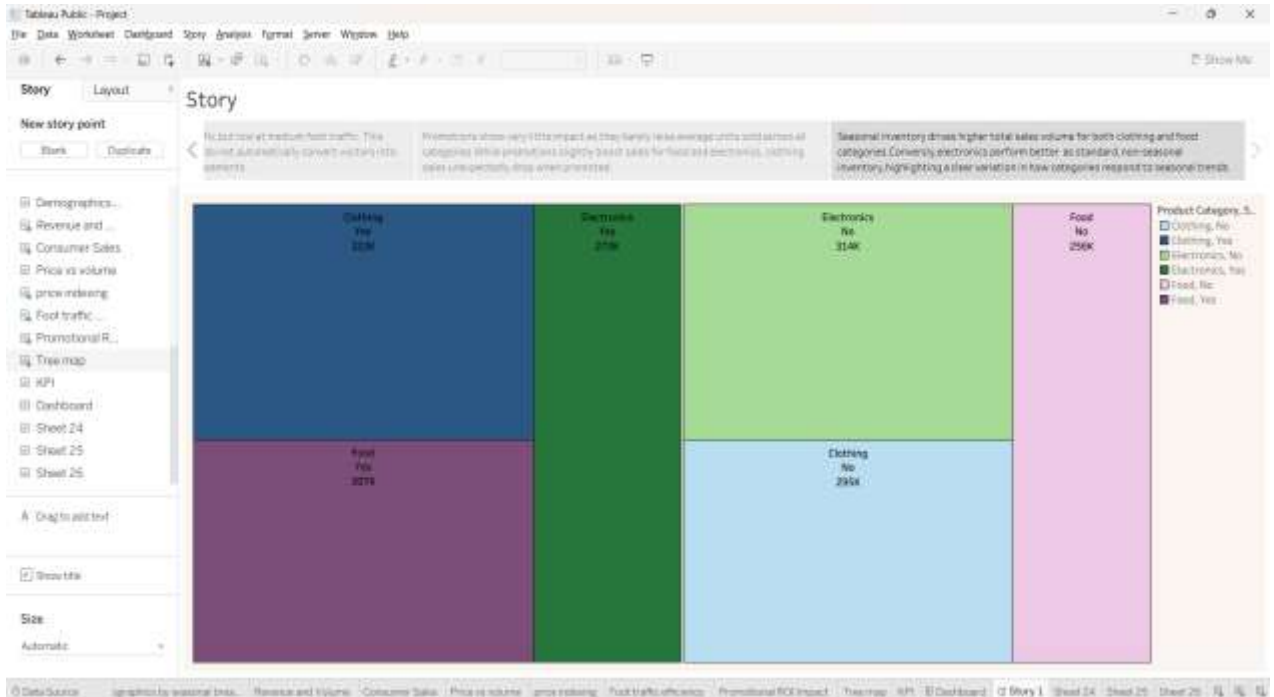
7. RESULTS

7.1 Output Screenshots









8. ADVANTAGES & DISADVANTAGES

Advantages:

- Enables data-driven business decision-making.
- Provides interactive and easy-to-understand dashboards.
- Optimizes product placement strategies.
- Improves understanding of consumer behavior.
- Supports analysis of sales, pricing, promotions, and seasonal trends.
- Saves time through automated visualization and reporting.
- Easy to use with Tableau's intuitive interface.
- Scalable for future business expansion.

Disadvantages:

- Dependent on the quality of the input dataset.
- Uses a static CSV dataset without real-time updates.
- Limited to the attributes available in the dataset.
- Does not provide predictive analytics.
- Requires data updates.
- Basic Tableau knowledge is required.
- Limited to product placement and sales analysis.

9. CONCLUSION

The Strategic Product Placement Analysis project demonstrates how Tableau can be used to analyze retail sales data and provide valuable insights into product placement strategies. By examining factors such as product positioning, product category, consumer demographics, sales volume, pricing, competitor pricing, promotions, seasonal trends, and foot traffic, the project identifies patterns that influence sales performance.

The interactive dashboards and visualizations enable users to understand complex data more effectively, compare different business metrics, and make informed decisions. The analysis helps retail companies optimize product placement, improve marketing strategies, enhance customer experience, and maximize sales opportunities.

Overall, this project highlights the importance of data visualization and business intelligence in the retail sector. It provides a practical, user-friendly solution that supports data-driven decision-making and demonstrates how Tableau can transform raw retail data into actionable business insights. With future enhancements such as real-time data integration and predictive analytics, the solution can become even more effective in supporting retail business growth.

10. FUTURE SCOPE

The future scope of this project includes integrating real-time retail data, implementing predictive analytics, enhancing customer segmentation, supporting multi-store analysis, automating report generation, and developing mobile-friendly dashboards. These enhancements will provide deeper business insights, improve decision-making, and help retail companies optimize product placement strategies more effectively.

11. APPENDIX

Source Code(if any)

Dataset Link:

<https://www.kaggle.com/datasets/amitvkulkarni/impact-of-product-positioning-on-sales>

GitHub & Project Demo Link

Demo Link:

https://drive.google.com/file/d/1ru3J_BJQguFW5U_NEYHMhCTg8NPHVkn8/view?usp=sharing